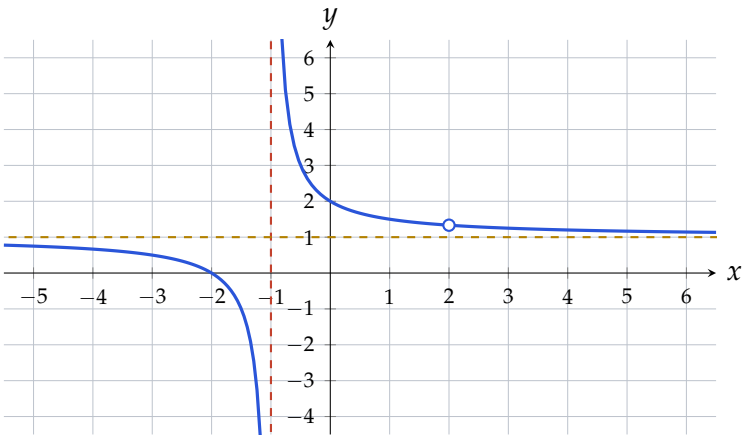
Lesson 1-3 • Limits to Infinity and Horizontal Asymptotes

Notes (filled in)

Objective: (1) find vertical asymptotes and holes; (2) use the top/bottom rules for limits to infinity and horizontal asymptotes.

The graph	 Graph of $f(x) = \dfrac{x^2-4}{x^2-x-2}$
Vertical asymptotes (VA)	Factor: $f(x) = \dfrac{x^2-4}{x^2-x-2} =$ $\dfrac{(x-2)(x+2)}{(x-2)(x+1)}$ **Vertical asymptote**: the denominator is 0 after cancelling: $x=-1$. **Hole**: a factor that cancels: $x=2$, at height $\frac{2+2}{2+1} = \frac{4}{3}$. A hole is a removable discontinuity. Near a VA the limit is $\pm\infty$: $\lim_{x\to -1^+} f(x) = +\infty$, $\lim_{x\to -1^-} f(x) = -\infty$.
Horizontal asymptotes (HA)	A horizontal asymptote comes from a limit to infinity: $\lim_{x\to\infty} f(x) = 1$, so the HA is $y=1$. Fun fact: for a rational function with a HA, $\lim_{x\to\infty} f(x) = \lim_{x\to -\infty} f(x)$: the same HA on both ends.
Limits to infinity: top/bottom rules	Compare the highest powers (degrees) of the top and the bottom. 1) $\lim_{x\to\pm\infty} \frac{\text{BIG}}{\text{small}} =$ $\pm\infty$ (DNE, no HA) bigger degree on top 2) $\lim_{x\to\pm\infty} \frac{\text{small}}{\text{BIG}} =$ 0 (HA $y=0$) bigger degree on the bottom 3) $\lim_{x\to\pm\infty} \frac{\text{same}}{\text{same}} =$ ratio of the leading coefficients same degree

Examples	1) $\lim_{x\to\infty} \dfrac{4x^3 - x}{5 - 2x^2} =$ DNE $(-\infty)$ 2) $\lim_{x\to-\infty} \dfrac{3x}{x^2+7} =$ 0 3) $\lim_{x\to\infty} \dfrac{6x^2+1}{3-2x^2} =$ $\dfrac{6}{-2} = -3$ 1) degree $3 > 2$: grows without bound; $\frac{4x^3}{-2x^2} = -2x \to -\infty$. 2) degree $1 < 2$: 0. 3) same degree 2: $\frac{6}{-2} = -3$; HA $y = -3$.
Summary	Factor first: a cancelled factor gives a hole, a zero left in the denominator gives a vertical asymptote. For $x \to \pm\infty$, compare degrees: top bigger $\to \pm\infty$, bottom bigger $\to 0$, equal $\to$ ratio of leading coefficients (the HA).